

Assignment: Caches

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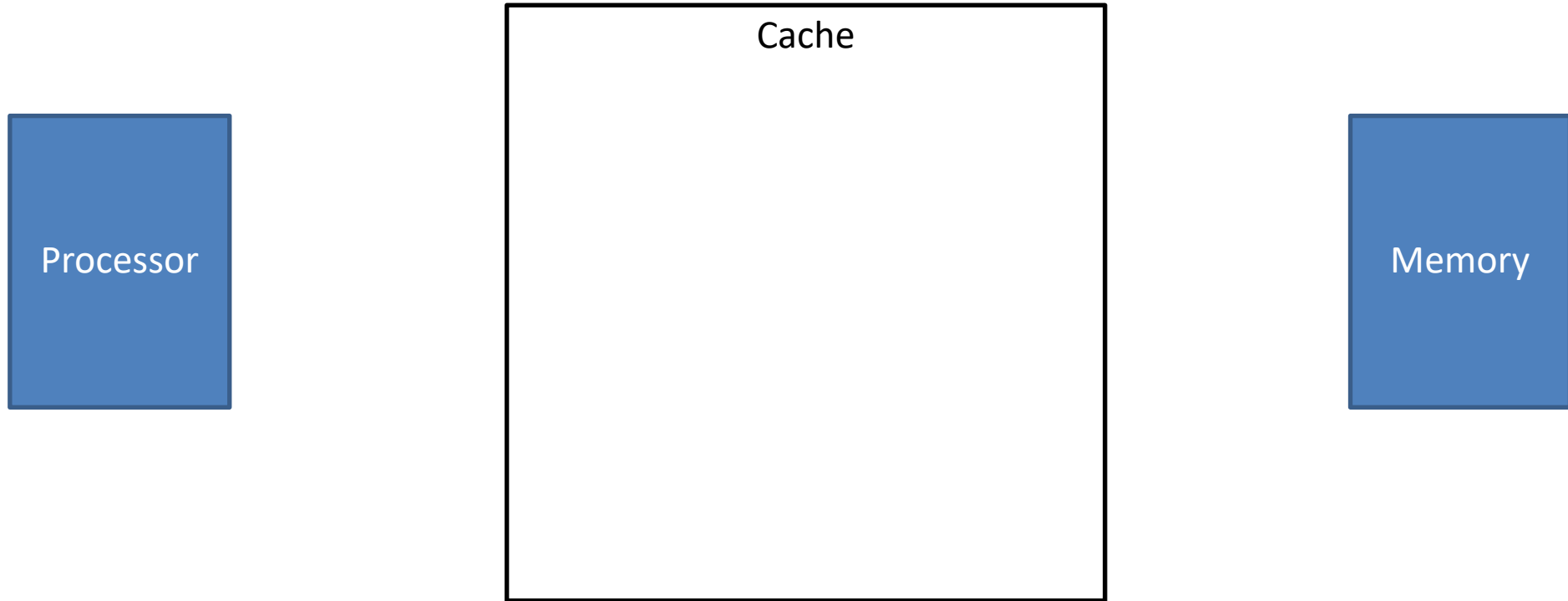
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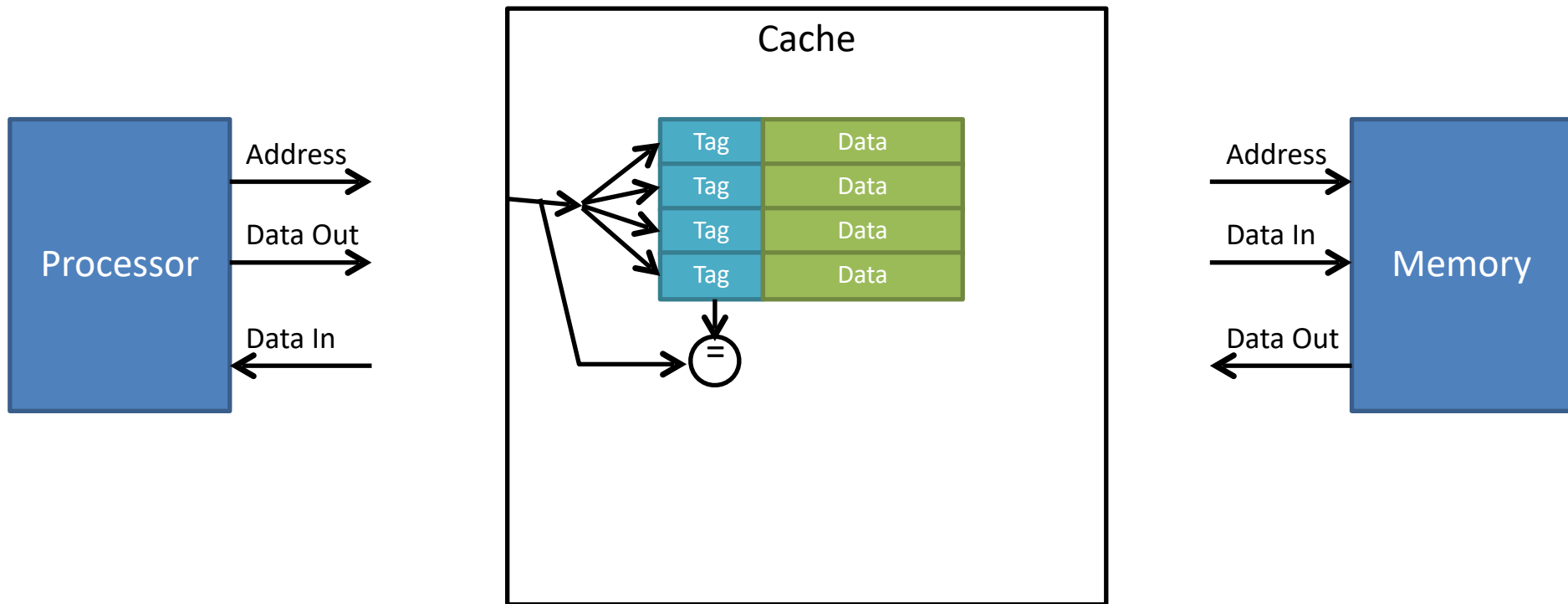
Grading (total 4 points)

- Question 1: 1 point
 - Part B: 0.4 points, Part C: 0.4 points
- Question 2: 1.2 point
 - Each part 0.4 point
- Question 3: 1.4 point
 - Part A, B, C each 0.4 point, Part D 0.2 point
- Question 4: 0.6 point
 - Each part 0.2 point

Review: What goes into and out from a cache?

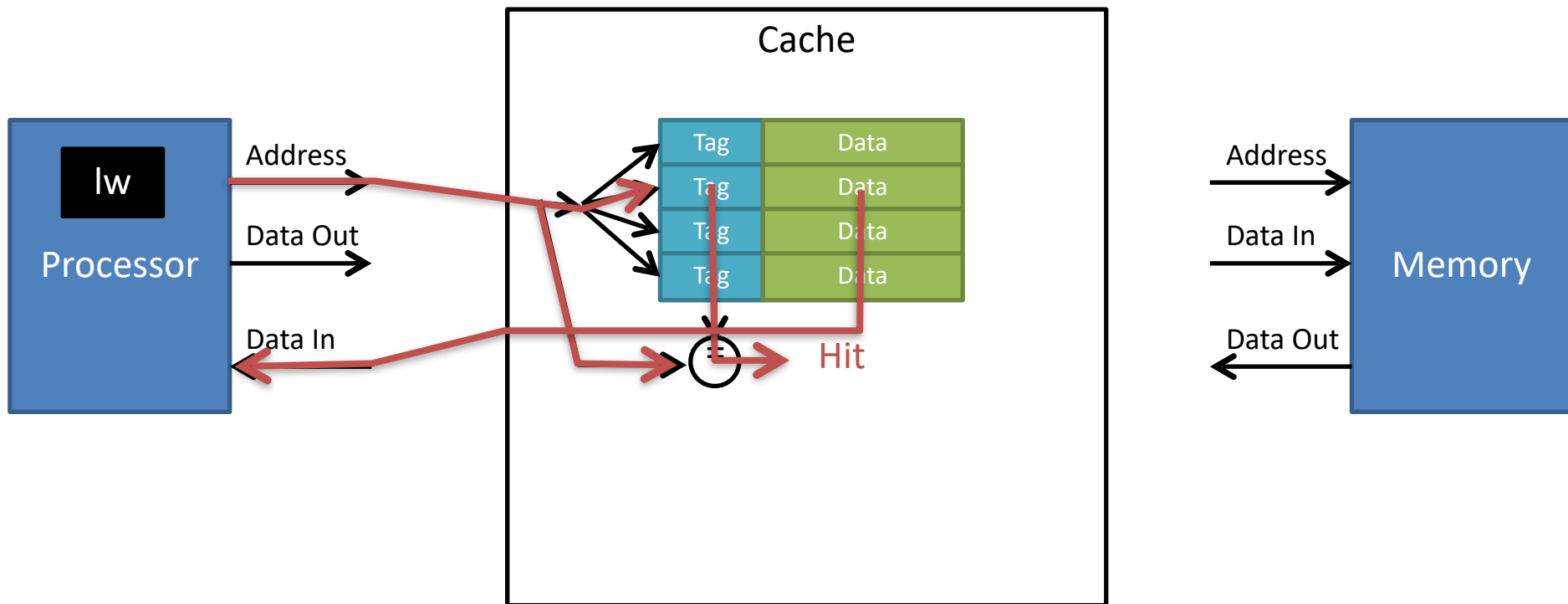


What goes into and out from a cache?



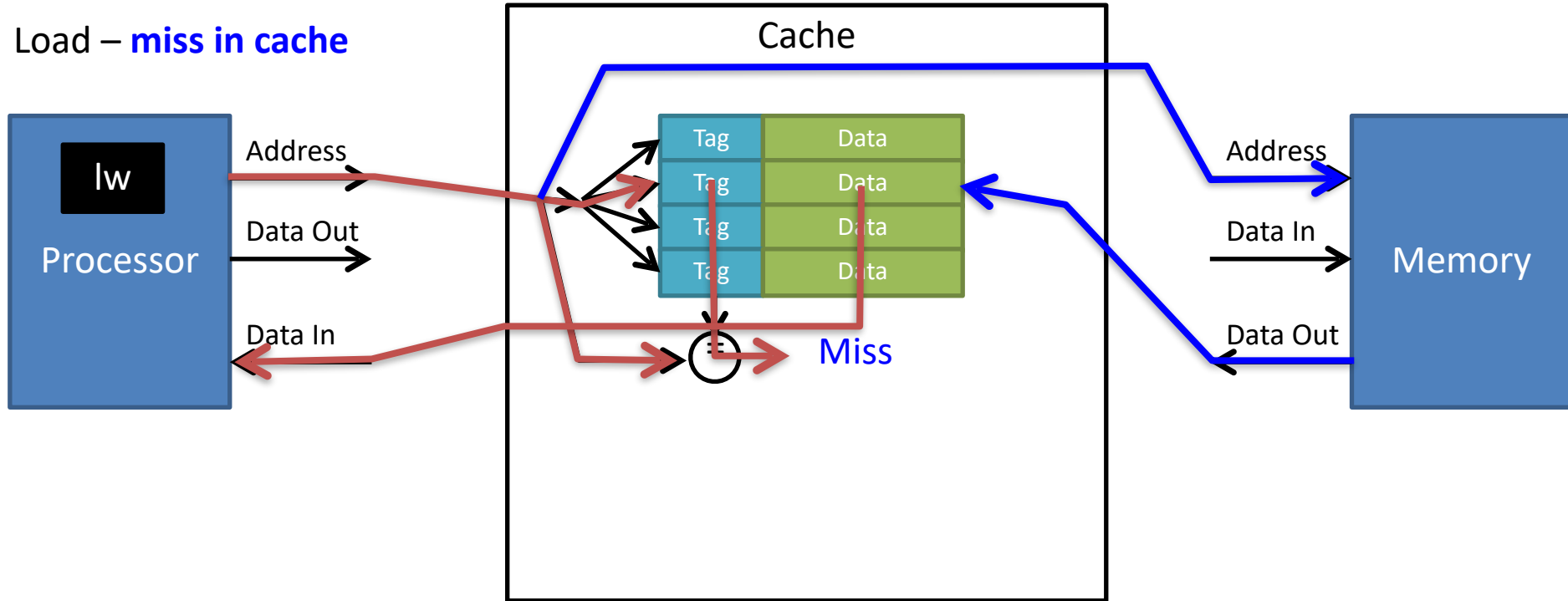
What goes into and out from a cache?

Load – **hit in cache**



What goes into and out from a cache?

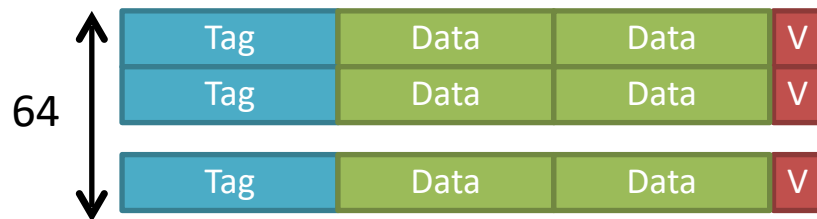
Load – **miss in cache**



Question 1: Diagram of the Caches

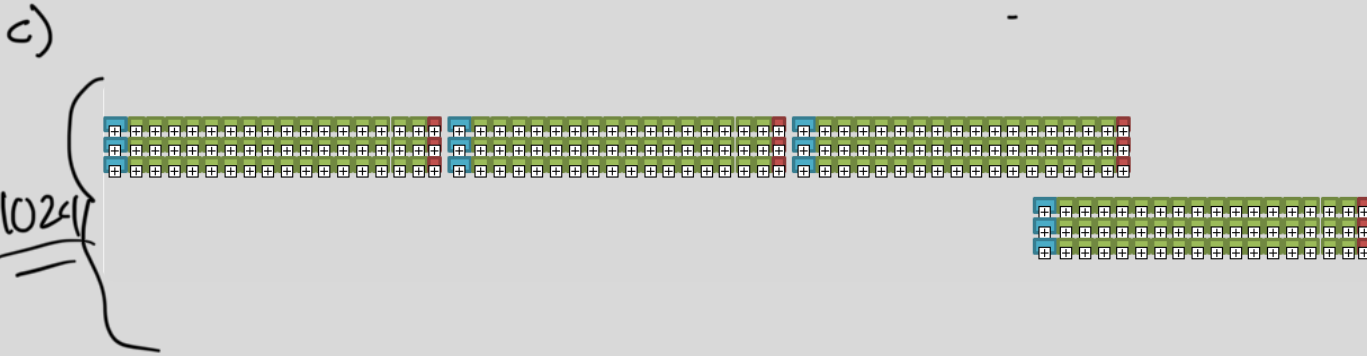
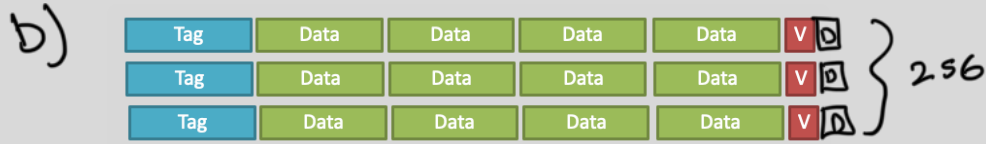
Draw a diagram of the following caches (note: answer for part A is provided. Follow the same structure for part B and C)

- A. Direct-mapped, cache capacity = 64 cache line, cache line size = 8-byte, word size = 4-byte, write strategy: write through



- B. Fully-associative, cache capacity = 256 cache line, cache line size = 16-byte, word size = 4-byte, write strategy: write back
- C. 4-way set-associative, cache capacity = 4096 cache line, cache line size = 64-byte, word size = 4-byte, write strategy: write back

Answer



Question 2: How are the bits used for addressing?

- Determine which bits in a 32-bit address are used for selecting the **byte (B)**, selecting the **word (W)**, indexing the **cache (I)**, and the cache **tag (T)**, for each of the following caches:

- Direct-mapped, cache capacity = 64 cache line, cache line size = 8-byte, word size = 4-byte
- Fully-associative, cache capacity = 256 cache line, cache line size = 16-byte, word size = 4-byte
- 4-way set-associative, cache capacity = 4096 cache line, cache line size = 64-byte, word size = 4-byte

Note: cache capacity represents the maximum number of cache blocks (or cache lines) that can fit in the cache

Answer

- a) Byte = 2 Bit, word = 1-bit, cache = 6-bit, tag = 23-bit
- b) Byte = 2 bit, Word = 2 bit, cache = 0-bit, tag = 28-bit
- c) Byte = 2 Bit, word = 4 bit, cache = 10-bit, tag = 16-bit

Question 3: How much overhead for the caches?

- Calculate the number of bits used for actual data and overhead (i.e., tag, valid bit, dirty bit) for each of the following caches
- A. Direct-mapped, cache capacity = 64 cache line, cache line size = 8-byte, word size = 4-byte, **write strategy: write through**
- B. Fully-associative, cache capacity = 256 cache line, cache line size = 16-byte, word size = 4-byte, **write strategy: write back**
- C. 4-way set-associative, cache capacity = 4096 cache line, cache line size = 64-byte, word size = 4-byte, **write strategy: write back**
- D. How the cache line size affect the overhead?

Answer

a) Data: $64 \text{ lines} * (8 \text{ bytes/line}) = 512 \text{ bytes} = 4096 \text{ bits}$

overhead: $64 \text{ lines} * (23 \text{ bits/tag/line} + 1 \text{ valid/line}) = 1536 \text{ bits}$

73% data

b) Data: $256 \text{ lines} * (16 \text{ bytes/line}) = 4096 \text{ bytes} = 32768 \text{ bits}$

overhead: $256 \text{ lines} * (28 \text{ bits/tag/line} + 1 \text{ valid bit/line} + 1 \text{ dirty/line}) = 7680 \text{ bits}$

81% data

c) Data: $4096 \text{ lines} * (64 \text{ bytes/line}) = 262144 \text{ bytes} = 2097152 \text{ bits}$

overhead: $4096 \text{ lines} * (16 \text{ bits/tag/line} + 1 \text{ valid bit/line} + 1 \text{ dirty bit/line}) = 73728 \text{ bits}$

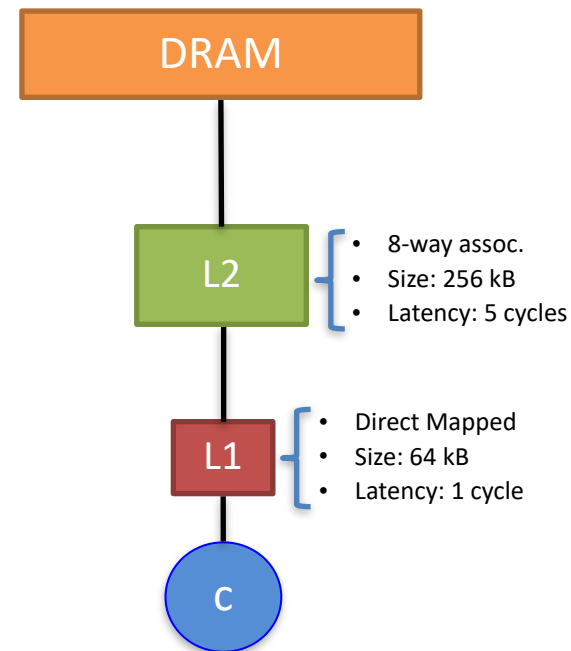
97% data

d) Typically, larger cache lines reduce overhead

Question 4: Average Memory Access Time

Consider the processor (on right) with two-level cache hierarchy. Suppose that **65%** of *benchmark A*'s instructions are accessed from the **L1 cache**, whereas the rest come from the **L2 cache**.

- What is the average access time for memory instructions in the benchmark
- If we change the L1 cache to be 4-way set associative we increase its flexibility and find that 85% of the instructions are now accessed from the L1. What is the average access time for memory instructions in the benchmark now?
- What is the speedup for changing the L1?



Answer

$$a) (0.65 * 1 \text{ cycle}) + 0.35 * (5 \text{ cycles} + 1 \text{ cycle}) = \underline{\underline{2.75 \text{ cycle}}}$$

$$b) (0.85 * 1 \text{ cycle}) + 0.15 * (5 \text{ cycle} + 1 \text{ cycle}) = \underline{\underline{1.75 \text{ cycle}}}$$

$$c) \underline{\underline{2.75 / 1.75 \approx 1.57 \text{ times faster so, about 36\% faster}}}$$